

all three rows are collinear; $M(H)$ singles out the ferromagnet, and only the spin splitting $\Delta(\mathbf{k}) = E_\uparrow(\mathbf{k}) - E_\downarrow(\mathbf{k})$ separates the other two

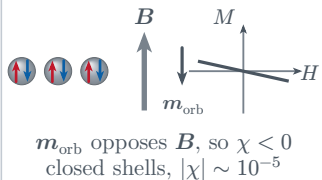
real-space structure

response $M(H)$

spin-resolved bands

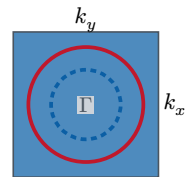
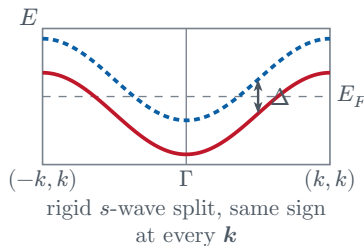
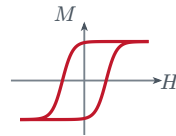
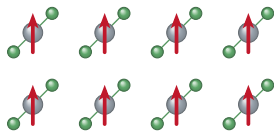
spin splitting $\Delta(\mathbf{k})$

Diamagnetism

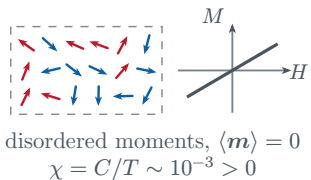


Ferromagnetism

one sublattice, every moment parallel

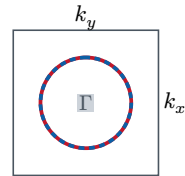
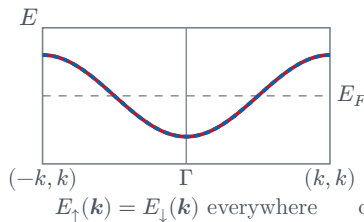
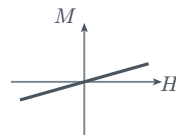
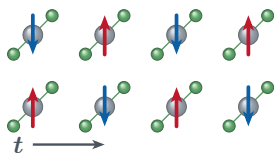


Paramagnetism

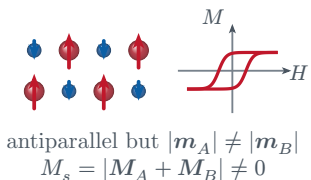


Antiferromagnetism

sublattices related by a translation $\mathbf{t}$, or by inversion

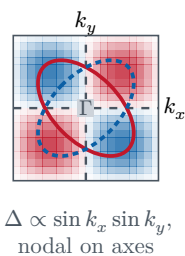
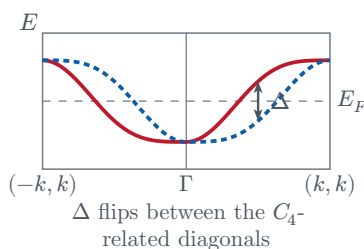
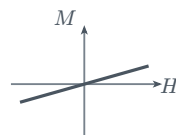
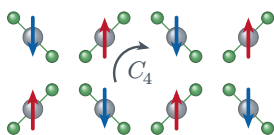


Ferrimagnetism



Altermagnetism

sublattices related only by a C_4 rotation



— $E_\uparrow$ — $E_\downarrow$ curves drawn on top of each other are spin degenerate ■ $\Delta > 0$ ■ $\Delta < 0$

non-relativistic limit throughout; $M(H)$ axes not to scale across panels; d -wave altermagnet shown, g - and i -wave also exist